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E-mail:xbywb@ouc.edu.cn

Comparison of Burrowing-Out Performance and Efficiency Between Dual-Anchor and Extension-Contraction Soft Robots

HUANG Xin[1)], HE Jia[1), *], WANG Hao[1)], YAN Fengyuan[1)], and KOU Hailei[2)]

1) *Key Laboratory of Ministry of Education for Geomechanics and Embankment Engineering*, *Hohai University*, *Nanjing* 210098, *China*
2) *College of Engineering*, *Ocean University of China*, *Qingdao* 266404, *China*

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Abstract In this research, a comparative analysis was conducted on the performance and efficiency of the dual-anchor soft robot (DASR) and the extension-contraction soft robot (ECSR). These robots were constructed by imitating the locomotion of razor clams. The penetration force for extension actuators and the anchorage force for expansion actuators in dry sand with distinct relative densities were tested by differentiating input air pressure and length-to-diameter ratios (λ). On the basis of the findings, a DASR and an ECSR were developed. DASR comprised two expansion actuators as the head and the tail segments at two ends, and one extension actuator as the middle segment. ECSR was composed of an extension actuator. A method based on the force equilibrium was introduced to ascertain and adjust the geometric parameters (length of each segment) of DASR. The burrowing-out performance and efficiency of DASR and ECSR in sands with distinct relative densities were explored. The results revealed that DASR exhibited high efficiency in dense sand in terms of lower time of burrowing-out, slip-to-advancement ratio, and cost of transport. ECSR might perform better in looser sand in terms of higher average burrowing-out velocity, higher advancement in each cycle, and lower energy consumption. However, it had larger slips than DASR. DASR could realize steady advancement and net displacement in each cycle and effectively decrease slips. These findings demonstrate the benefits and usability of the dual-anchor motion and offer new insights into the application of the dual-anchor mechanism in the burrowing of robots.

Key words razor clam; soft robots; self-burrowing; sand; soil investigation

1 Introduction

During the last decade, subterranean robots have attracted increasing attention (Jacobstein *et al*., 2017; Astley *et al*., 2020; Naclerio *et al*., 2021; Wei *et al*., 2021; Martinez *et al*., 2022; Zhang *et al*., 2022a; Zhang *et al*., 2022b; Zhang *et al*., 2023a; Zhang *et al*., 2023c). These robots can perform tasks in dangerous or extreme environments, including geotechnical surveys on mountainous terrains (Martinez *et al*., 2022; Zhang *et al*., 2024b), subsurface sensor installations (Martinez *et al*., 2022), seabed prospecting (Zhang *et al*., 2022a), explorations in extremely contaminated environments (Zhong and Tao, 2022), post-disaster search and rescue (Naclerio *et al*., 2021), and extraterrestrial investigations (Wei *et al*., 2021). The gradual expansion of construction and development into complex and extreme environments has required geological exploration and monitoring technologies to be flexible, efficient, cost-effective, and adaptable to complex environments (Martinez *et al*., 2022; Zhang *et al*., 2023b; Zhang *et al*., 2024b). At present, *in situ* geological and geotechnical testing generally depends on field tests, including the cone penetration test (CPT), field vane test, and flat dilatometer test (DMT). In areas with restricted site conditions encompassing underwater, steep slopes, lakes, and marshes, these tests cannot generate sufficient reaction force to reach the desired depth (Martinez *et al*., 2020). Moreover, these tests can typically only obtain soil parameters at discrete points in the subsurface or continuous soil parameters in the vertical direction. Assessing soil properties and gathering geological information along horizontal directions are usually difficult. Subterranean robots could be an alternative method for new geological investigation and monitoring.

The locomotion of robots in the subsurface is relatively complex. The resistance to move forward in soil is one to two orders of magnitude greater than that in air and water (Tao, 2021). The inherent vertical stress gradient in soil generates a lift force and tends to tilt upward when moving horizontally (Ding *et al*., 2011). A large number of animals, plants, and microorganisms, including insects, mollusks, fish, bacteria, fungi, and plant roots, are present in subsurface and near-surface environments (Dorgan, 2015; Martinez *et al*., 2022; Zhang *et al*., 2023b; Zhang *et al*., 2024b). These organisms adapt to and interact with

* Corresponding author. E-mail: hejia@hhu.edu.cn

the geological environment by utilizing different types of functions encompassing movement, growth, anchoring, and nutrient uptake in the soil (Dorgan, 2015; Hosoi and Goldman, 2015; Dorgan and Daltorio, 2023; Riiska *et al*., 2024). These biological activities can also impact and vary the properties of the geological environments (Martinez *et al*., 2022). Soil organisms realize locomotion, feeding, and reproduction with the help of locomotion strategies that are efficient, flexible, safe, and low energy consuming, such as peristalsis of earthworms (Dorgan, 2015, 2018), dual-anchor of bivalves (Trueman, 1967; Winter *et al*., 2012; Pirrone *et al*., 2024), tip extension and circumnutation of roots (Mullen *et al*., 1998; Ruiz *et al*., 2015; Del Dottore *et al*., 2017), undulatory swimming of sandfish lizard (Baumgartner *et al*., 2008; Maladen *et al*., 2009; Ding *et al*., 2012), sand fluidization of octopuses (Montana *et al*., 2015), rotation during self-burial of seeds (Jung *et al*., 2017), digging of mole-rats (Van Wassenbergh *et al*., 2017), burrowing of sand crabs (Trueman, 1970; Treers *et al*., 2022), vibration (Cong *et al*., 2021; Dong and Gong, 2024), locust ovipositor valves (Das *et al*., 2022), and ovipositor of woody wasps (Alkalla *et al*., 2021; Alkalla and Pitcher, 2024). In general, these biological strategies can be categorized into three: leaving a cave after locomotion (*e.g*., digging of mole-rats), movement of organisms facilitated by interaction with soil (*e.g*., tip extension of roots, dual-anchor of bivalves, undulatory swimming of sandfish lizard, and peristalsis of earthworms), and assisted resistance reduction strategies (*e.g*., sand fluidization of octopuses and rotation during self-burial of seeds). The evolutionary modification of these locomotion strategies over millions of years has been rigorously confirmed through natural selection (Martinez *et al*., 2022; Sun *et al*., 2023). Bioinspired robots implementing such biomechanically optimized principles exhibit superior environmental adaptability and terrain negotiation capabilities to conventional engineering solutions, specifically in complex subterranean environments (Naclerio *et al*., 2021).

Recently, several preliminary subterranean robots have been constructed based on the locomotion strategies of soil organisms. These robots include traditional rigid robots, rigid-flexible coupling robots, and soft robots. The subterranean locomotion (horizontal or vertical) tests of these robots have utilized soils that are predominantly dry granular materials (glass beads, plastic beads, and sand). Chopra *et al*. (2023) designed a rigid robot, which was inspired by the underactuated appendages of sea turtles, to enable horizontal movement in shallow soils (depth of approximately 13 cm). Kim *et al*. (2023) presented a reciprocating horizontal burrower using foldable origami feet, which could provide variable resistances for either moving forward or anchoring in soil. The burrower realized a reciprocating horizontal motion in shallow granular media (5 cm) by using a horizontal support frame.

With regard to vertical rigid subterranean robots, Treers *et al*. (2022) presented the EMerita BUrrowing Robot (EMBUR) based on the burrowing strategies of sand crabs. EMBUR could move obliquely from the surface of the plastic bead to the subsurface with a depth of 2 cm because of its retractable legs. Some vertical penetrating rigid robots have also developed based on localized fluidization of razor clams and seed rotation (Winter *et al*., 2014; Bagheri *et al*., 2024). These robots need to be preburied in the soil to provide sufficient reaction forces and could reach a maximum depth of nearly 30 cm. A few researchers have designed earthworm-inspired rigid-flexible coupled peristaltic robots and probes by adopting a flexible membrane pressurized expansion as the expansion segment and a rigid device as the extension segment (Isaka *et al*., 2019; Naziri *et al*., 2024).

Rigid subterranean robots may encounter operational limitations in soil environments due to their inherent structural rigidity. Thus, the use of auxiliary equipment is frequently required for the navigation and functionality maintenance of these robots. Conventional, manually operated probes are widely implemented, but they usually entail elevated operational risks and suboptimal efficiency stemming from their operator-dependent manipulation paradigms. Soft robots are theoretically characterized by infinite degrees of freedom, which enable them to exhibit enhanced flexibility and operational safety coupled with exceptional adaptability to unstructured environments (Rus and Tolley, 2015; Ilami *et al*., 2021; Li *et al*., 2021). Furthermore, their biomimetic actuation mechanisms, which are analogous to biological locomotion principles, make them promising candidates for developing next-generation bioinspired subterranean exploration systems.

Das *et al*. (2023) designed and developed a modular soft robot inspired by the peristaltic mechanism. The robot was constructed based on a peristaltic soft actuator. Niiyama *et al*. (2022) built an earthworm robot by connecting shape memory alloy-driven segments in series. These robots can realize horizontal movement within a tube or through shallow sand. Soft robots primarily apply actuators driven by fluids, such as fiber-reinforced pneumatic actuators (Rus and Tolley, 2015; Connolly *et al*., 2017). By varying the angle of fiber winding, the actuators exhibit different deformation patterns, including radial expansion, axial extension, and twisting, as the fluid interacts with the inner surfaces of the actuators (Connolly *et al*., 2015). Tao *et al*. (2020) designed a minimalistic soft self-burrowing-out robot (SBOR) using a fiber-reinforced pneumatic actuator. SBOR could burrow out in dry and saturated sands. The aforementioned robots are devised based on a single locomotion strategy of organisms in the soil, and some other researchers have designed subterranean robots by combining multiple locomotion strategies of organisms (Ortiz *et al*., 2019; Naclerio *et al*., 2021; Tang *et al*., 2024; Zhang *et al*., 2024a). These robots use tip extension, dual anchors, or extension-contraction as the major driving strategy, as well as various assisted resistance reduction strategies (*e.g*., sand fluidization, tip rotation, and bi-directional bending). Most of these robots could realize horizontal movement in shallow granular media.

In this work, a dual-anchor soft robot (DASR) and an extension-contraction soft robot (ECSR) were developed by imitating the movements of razor clams. DASR com-

prised three fiber-reinforced pneumatic actuators: a head expansion actuator, a tail expansion actuator, and a middle extension actuator. ECSR was composed of one extension actuator that could perform single extension contraction. The performance of the actuators was adjusted under different input pressures. The effects of the length-to-diameter ratio and the density level of sand on the anchoring and penetration forces of a single actuator were assessed to optimize the size of the robots. In the sand with distinct relative densities (30%, 50%, and 80%), DASR and ECSR were utilized for burrowing-out tests. Typical burrowing curves were derived to examine the differences in burrowing-out performance. The average speed, energy consumption, and cost of transport (COT) of DASR and ECSR were comparatively investigated.

2 Design of the Soft Robots

2.1 Locomotion Strategies of Razor Clams

Razor clams, as a typical burrowing organism, differ in their downward and upward locomotion strategies (Fig.1) (Trueman, 1967; Tao *et al*., 2020; Tang *et al*., 2024). Razor clams use a double-anchored mechanism for downward movement but rely on direct telescoping for upward movement. A cycle of downward dual-anchor locomotion can be streamlined into four steps: first, the shell is opened to establish a shell anchorage. Secondly, the foot extends downward to penetrate the soil body, forming a tension crack. Thirdly, the shell closes, which extends the foot to produce a foot anchorage. Finally, the shell is closed, and the body is pulled downward to realize pure locomotion. A natural vertical stress gradient is present in natural soil. Therefore, razor clams can conduct simple extensions to move upward. During the upward movement, the upward advancement is greater than the downward slide during each cycle of locomotion, which facilitates net upward movement.

2.2 Fabrication of Fiber-Reinforced Pneumatic Actuators

In many works, soft actuators adopt Dragon Skin series silicone (Connolly *et al*., 2015; Rafsanjani *et al*., 2018; Pourghasemi Hanza and Ghafarirad, 2023). It has high strength and large elastic modulus and can withstand high air pressure. The silicone has adequate setting time for defoaming, pouring, and other operations throughout the production process. The actuator comprises an inner silicone layer, fibers, and an outer silicone layer. In this study, Dragon Skin 30 was used for the inner silicone layer, and Dragon Skin 10 medium was utilized for the outer silicone layer. The fiber-reinforced pneumatic actuators were fabricated following the method of Connolly *et al*.

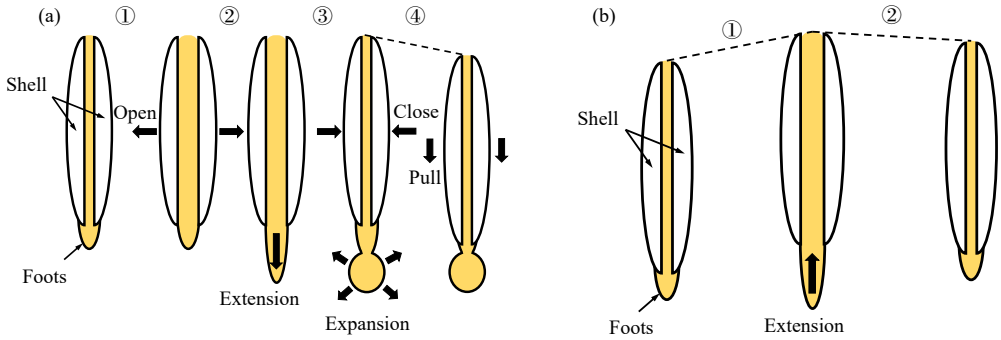

Fig.1 Burrowing strategy of the razor clam. (a), downward dual-anchor locomotion; (b), upward extension-contraction locomotion (adapted from Tang *et al*., 2024).

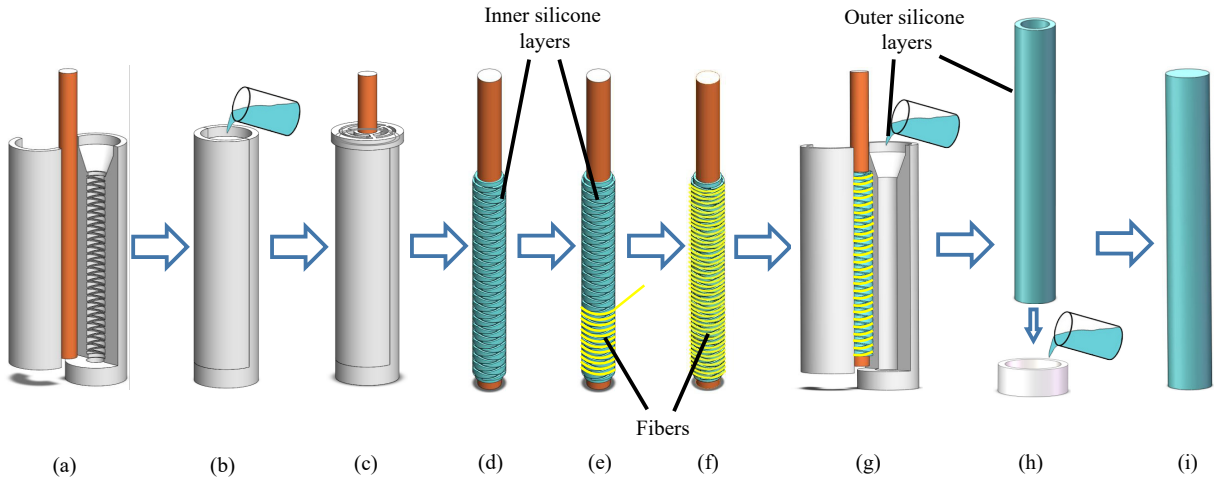

Fig.2 Fabrication of the soft pneumatic probe. (a), 3D-printed mold; (b), silicone pouring; (c), acrylic rod insertion; (d), setting completion; (e), fiber winding; (f), winding completion; (g), pouring of the outer silicone layer; (h), bottom sealing; (i), final actuator formation.

(2015) (Fig.2). The steps are as follows:

1) The inner silicone layer was poured using a 3D-printed mold comprising two parts with raised threads on the inside of the mold for fiber winding.

2) After the release agent was sprayed on the surface of the mold, the two parts of the mold were closely attached and bonded using glue to avoid leakage. The prepared silicone was poured into the mold until it was nearly three-fifths of the height of the mold. A cylindrical acrylic rod with a diameter of 15 mm was inserted into the mold to establish the inner cavity of the pneumatic actuators.

3) After the setting of the silicone, the fibers (Kevlar wire) were wrapped along the grooves of the silicone at an angle of ±20° relative to the axial lines for the expansion actuator and ±85° relative to the axial lines for the extension actuator. Once the fiber winding was complete, the outer silicone layer was situated using another set of molds in the same process as the inner silicone layer.

4) The corresponding bottom and top sealing molds were adopted to cover the two ends of each segment. One or more holes were created in some of the bottom/top covers for air tube installation in the assemblage step.

2.3 Control of the Fiber-Reinforced Pneumatic Actuators

The control system of the actuators consulted the Soft Robotics Toolkit by Holland *et al.* (2014). As shown in Fig.3, an air pump was utilized as the air source to deliver air flow to the solenoid valves. Solenoid valves controlled the inflation, holding, and deflation of an actuator. All solenoid valves and air pumps were controlled by an Arduino Mega 2560. Air pressure sensors were connected to the air circuit to monitor the real-time air pressure in the actuators for the adjustment of air pressure.

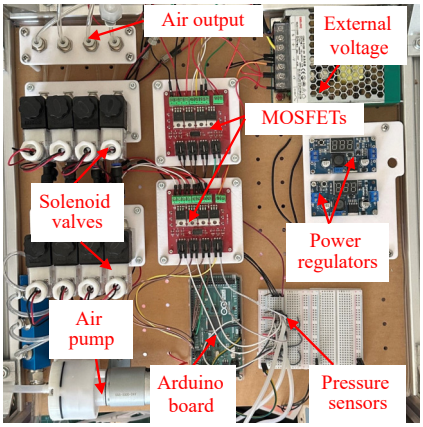

Fig.3 Control system for the soft robots.

2.4 Calibration of Fiber-Reinforced Pneumatic Actuators

By simply varying the fiber angle, the fiber-reinforced pneumatic actuators can realize a wide range of motions, such as axial extension, radial expansion, and twisting (Connolly *et al.*, 2015). This test evaluated the extension and expansion performance of the actuators. It also constructed an actuator without fiber reinforcement as a blank control for comparison with the extension and expansion actuators. The actuator was split into several segments (Fig.4). The extension ratio α and expansion ratio b in different input pressures were computed for each segment. The total extension ratio A and total expansion ratio B of the actuator subject to different pressures were also ascertained. The four parameters were computed to calibrate and assess the performance of the actuators. α, b, A, and B are defined as follows:

$$\alpha = \frac{l_m}{l}, \tag{1}$$

$$b = \frac{d_m}{d}, \tag{2}$$

$$A = \frac{L_m}{L}, \tag{3}$$

$$B = \frac{D_m}{D}, \tag{4}$$

where α is the extension ratio of the segment, l_m is the extended length of the segment, l is the initial length of the segment, b is the expansion ratio of the segment, d_m is the expanded diameter of the segment, and d is the initial diameter of the segment. A is the total extension ratio of the actuator, L_m is the extended length of the actuator, L is the initial length of the actuator, B is the total expansion ratio of the actuator, D_m is the expanded diameter of the actuator, and D is the initial diameter of the actuator.

The L and D of the actuator were 60 and 24 mm, respectively. The thickness (t) of the actuator was 4.5 mm. The thickness of the inner silicone layer was 2.5 mm, and the thickness of the outer silicone layer was 2 mm. According to our previous research, A for the expansion actuator and B for the extension actuator were maintained at approximately 1.0 (He *et al.*, 2024). When the limit input pressure p_m<130 kPa, the fluctuation of α in each segment is small (Fig.5a). When p_m=143 kPa and p_m=151

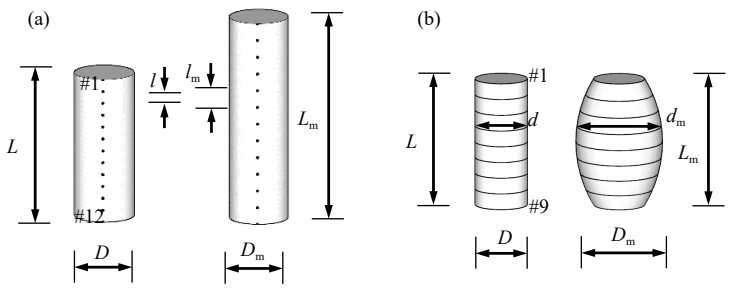

Fig.4 Extension and expansion of the actuators. (a), extension; (b), expansion.

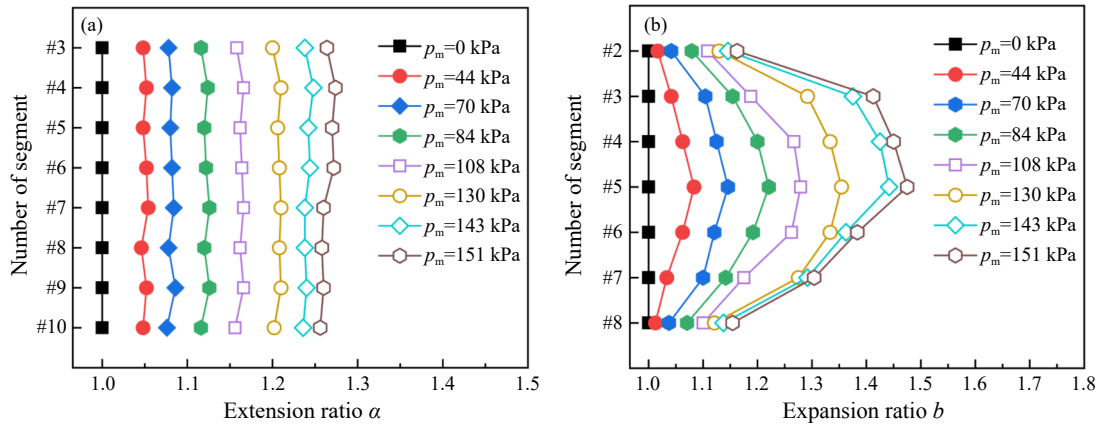

Fig.5 Extension and expansion ratio of the segmentssegments. (a), extension ratio α; (b), expansion ratio b.

kPa, the extension actuator does not produce complete axial extension, but it bends with a certain curvature, which results in the reduction in α in segments #6–#10. This phenomenon is also observed in the research of Connolly *et al*. (2015). The main reason may be that the axial extension of silicone raises the fiber gap, weakens the fiber constraint on the actuator, and causes the actuator to bend. The parameter b for the expansion actuator has a spindle shape with a large middle and two small sides, as shown in Fig.5b. When p_m<130 kPa, b is symmetrical. While p_m=143 kPa and p_m=151 kPa, the b values of segments #2–#5 are greater than those of segments #5–#8. The larger p_m amplifies the asymmetry in fiber winding. The optimal pressure p_m for extension and expansion actuators is within 130 kPa. When the input pressure exceeds 130 kPa, the extension actuator no longer exhibits uniform axial extension but bends due to asymmetric fiber constraints (Fig.5a). Similarly, the expansion actuator presents non-uniform expansion (Fig.5b), which reduces its anchorage effectiveness. Therefore, 130 kPa was selected as the upper limit to ensure stable and efficient actuator deformation.

We also tested the variation in A for extension actuators at different length-to-diameter ratios λ and the variation in b for expansion actuators. λ is defined as follows:

$$\lambda = \frac{L}{D}. \tag{5}$$

The D of the actuator was 24 mm, and the L was 36–84 mm. A is barely affected by λ when the values of λ are within 1.5–3.5 (Fig.6a). As shown in Fig.6b, b gradually increases as λ rises when $\lambda \leq 2.5$, and b is unchanged for λ=2.5 and λ=3.5.

2.5 Penetration and Anchorage Force of the Actuators

We tested the penetration force F_p of the extension actuators and the anchorage force F_a of the expansion actuators to determine the λ values of the actuators for the soft robots. Penetration force refers to the vertical force required for an actuator to push itself into the granular medium, which reflects its capability to advance through sand. Anchorage force is the vertical resistance force exerted by the surrounding sand when the actuator is pulled out after expansion, which ascertains its capacity to maintain a stable position during motion. A higher penetration force means better movement capability, while a higher anchorage force improves stability during burrowing. The D of the actuator was 24 mm, and the L was 36–84 mm. The λ values were from 1.5 to 3.5. The test was conducted on a desktop compression/tension machine (Fig.7a). The actuator was hooked to the rod of the compression/tension machine on one end and was preburied in a 1000 mL beaker filled with sand on the other end. The input air pressure was increased continuously while applying the penetration force to measure the pene-

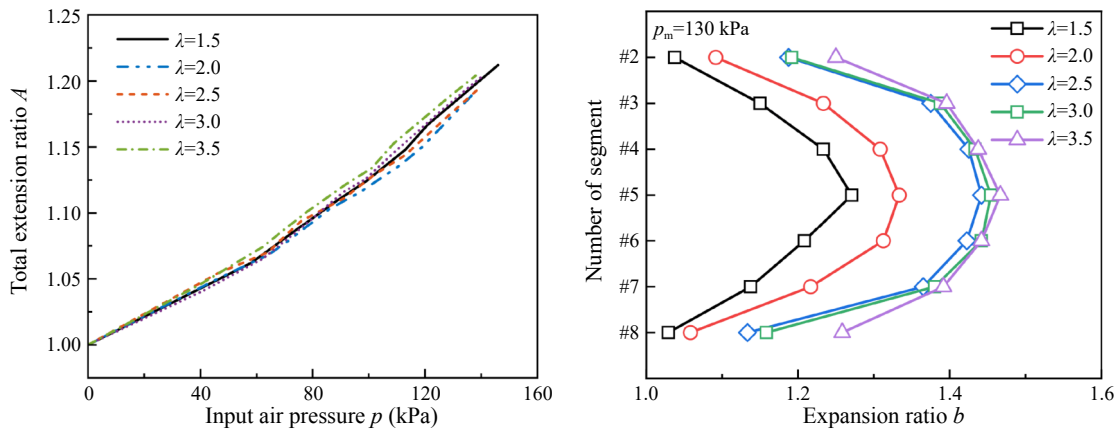

Fig.6 Effect of length-to-diameter ratio λ on extension and expansion performance. (a), total extension ratio A; (b), expansion ratio b.

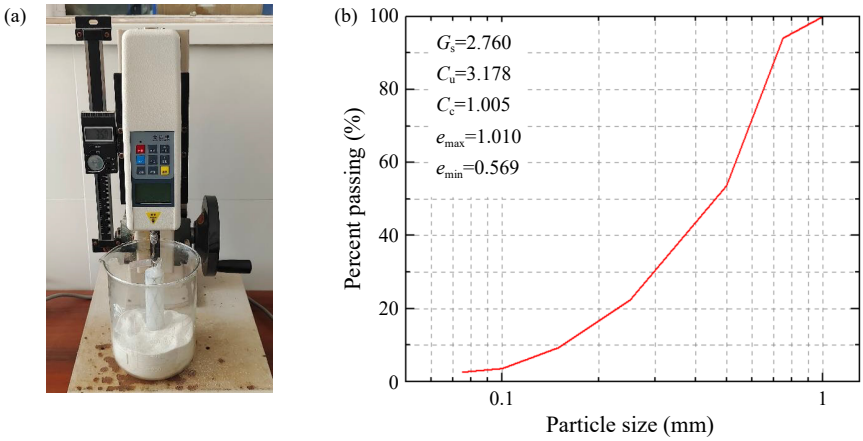

Fig.7 Penetration and anchorage force test of the actuators. (a), setup; (b), particle size distribution of sand.

tration force of the extension actuators. The anchorage force of the expansion actuators was measured by first increasing the input air pressure of the actuator to 130 kPa. Then, the actuator was pulled upward. The resistant force during pulling out was recorded. The mean particle size (D_{50}) of sand was 0.473 mm. The maximum void ratio e_{max} and minimum void ratio e_{min} of the sand were 1.010 and 0.569, respectively. The specific gravity G_s was 2.76. The coefficients of uniformity C_u and the coefficients of curvature C_c of the sand were 3.178 and 1.005, respectively. The particle size distribution of sand is shown in Fig.7b. The relative density (D_r) of the sand was 50% when testing the penetration force, and the relative densities of the sand were 30%, 50%, and 80% for testing the anchorage force. According to the American Society for Testing and Materials (ASTM) D2487-17 standard, the sand properties and relative density of the sand used in this study align with those commonly encountered in engineered fills and natural soils across practical engineering scenarios, such as structural backfill for building foundations, subgrade materials, and natural sand deposits.

The penetration force of extension actuators increases roughly linearly with the input air pressure and λ (Fig.8a). However, beyond λ=2.5, the additional increase in penetration force becomes less significant, which suggests diminishing returns on penetration efficiency. As shown in Fig.8b, the anchorage force of expansion actuators increases slightly with λ for D_r=30% and D_r=50%. However, at D_r=80%, a significant increase in anchorage force is observed when λ rises from 2.5 to 3.0. This result implies that a higher λ could improve anchoring in dense sand. However, increasing λ also results in longer actuators, which may weaken structural stability and raise control complexity. Considering the trade-offs among penetration, anchorage, and actuator stability, λ=2.5 was selected as the optimal ratio to balance these factors.

3 Burrowing-Out Test of the Dual-Anchor and Extension-Contraction Soft Robots

3.1 Burrowing-Out Strategies, Control, and Assemblage of the Soft Robots

3.1.1 Burrowing-out strategies and control

The burrowing-out strategies of the two soft robots were inspired by the razor clam. They both performed periodic cyclic burrowing. DASR comprised three fiber-reinforced pneumatic actuators: a head expansion actuator, an extension actuator, and a tail expansion actuator (Fig.9a). ECSR was composed of an extension actuator (Fig.9b). During a burrowing-out cycle, the tail expansion actuator of DASR was activated, followed by the ac-

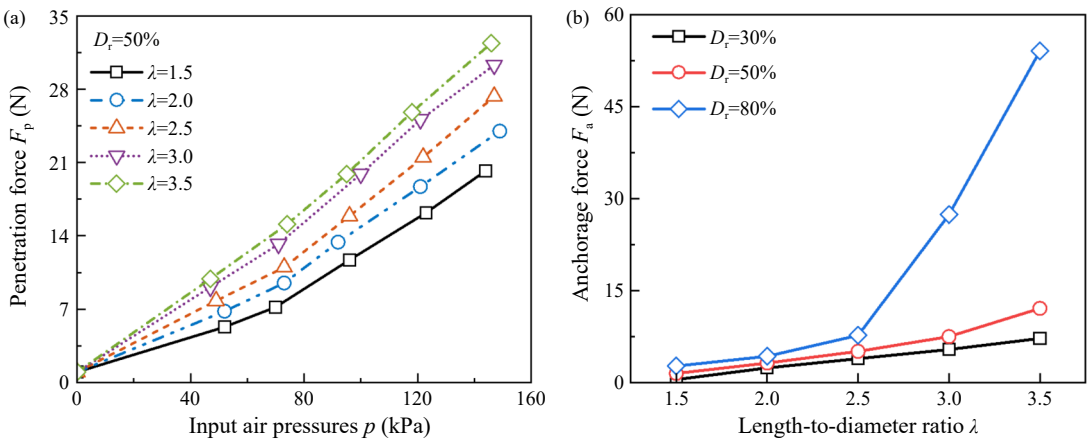

Fig.8 Effect of length-to-diameter ratio λ on penetration and anchorage force. (a), penetration force F_p; (b), anchorage force F_a.

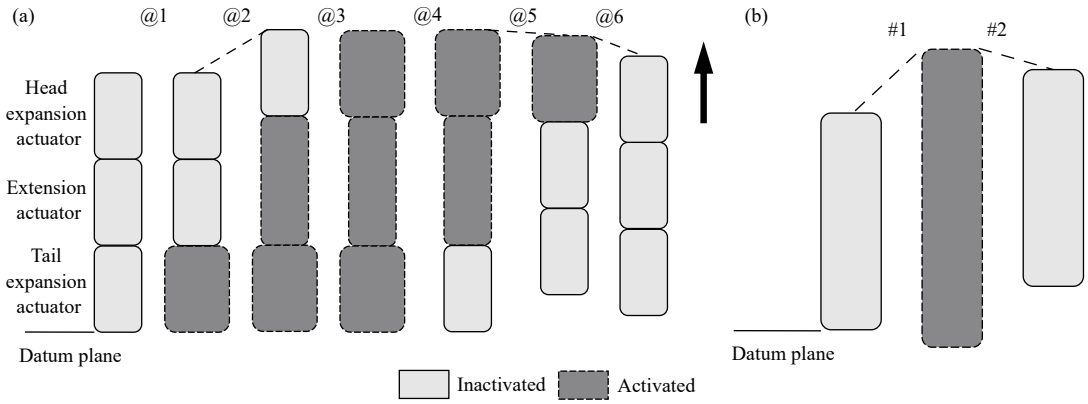

Fig.9 Burrowing-out strategies of the two soft robots. (a), dual-anchor soft robot; (b), extension-contraction soft robot.

tivation of the extension actuator for advancement. Then, the head expansion actuator was activated for anchorage. Finally, the tail expansion and the extension actuators were de-activated for the pull up of the robot body. The entire process is described in Fig.9a. For ECSR, the extension actuator was activated purely periodically (Fig.9b).

The soft robots were controlled by the control system described in Section 2.3. For DASR and ECSR, the pressure of each actuator was monitored separately using an air pressure sensor.

3.1.2 Geometric parameters of DASR

The geometric parameters of the DASR need to be determined to ensure its effective burrowing-out performance. The DASR has four unknown geometric parameters: the lengths of the head and tail expansion drivers (L_{d1} and L_{d3}), the length of the elongation driver (L_{d2}), and the diameter of the robot (D_p). We developed a mechanical model based on force equilibrium analysis to guide the design of the actuators. Specifically, the actuator must generate sufficient penetration and anchoring forces while maintaining structural stability during the burrowing-out cycle. This model was employed to determine the optimal geometric parameters of DASR, specifically the segment lengths, by analyzing the force balance in key gaits. Then, these equations were used to adjust the length-to-diameter ratio (λ), expansion actuator length (L_{d1} and L_{d3}), and extension actuator length (L_{d2}) to ensure efficient burrowing and minimal slip.

In each cycle, the movement of the DASR is primarily driven by gaits @2 and @5 (Fig.9a). Based on the force equilibrium in the two gaits (Fig.10), the following two equations can be derived:

$$\frac{\pi D_p^2}{4}\cdot q_{top,\,2}+\pi D_p(L_{d1}+L_{d2})\cdot\tau_{2h}+G=\pi D_{p,\,m}L_{d3}\cdot\tau_{target,\,2}+\frac{\pi D_{p,\,m}^2}{4}\cdot q_{t,\,2}, \tag{6}$$

$$\frac{\pi D_{p,\,m}^2}{4}\cdot q_{top,\,5}+\pi D_p(L_{d3}+L_{d2,\,m})\cdot\tau_{5t}+G=\pi D_{p,\,m}L_{d1}\cdot\tau_{target,\,5}+\frac{\pi D_p^2}{4}\cdot q_{t,\,5}, \tag{7}$$

where $q_{top,\,2}$ and $q_{t,\,2}$ represent the stress on the head and tail of the DASR from the sand during gait @2, respectively; τ_{2h} and $\tau_{target,\,2}$ stand for the downward and upward frictional stress from the sand during gait @2, respectively; $q_{top,\,5}$ and $q_{t,\,5}$ signify the stress on the head and tail of the DASR from the sand during gait @5, respectively; τ_{5t} and $\tau_{target,\,5}$ denote the downward and upward frictional stress from the sand during gait @5, respectively; $L_{d2,\,m}$ is the length of the extension actuator after extension; $D_{p,\,m}$ is the diameter of the expansion actuators after expansion; and G is the gravity.

τ_{2h}, $\tau_{target,\,2}$, τ_{5t}, and $\tau_{target,\,5}$ can be calculated using the following equations:

$$\tau_{2h}=\mu p_{2h}, \tag{8}$$

$$\tau_{target,\,2}=\mu p_{target,\,2}, \tag{9}$$

$$\tau_{5t}=\mu p_{5t}, \tag{10}$$

$$\tau_{target,\,5}=\mu p_{target,\,5}, \tag{11}$$

where p_{2h} and $p_{target,\,2}$ represent the horizontal stresses in the sand on the non-expansion actuators and tail expansion actuator during gait @2, respectively. p_{5t} and $p_{target,\,5}$ denote the horizontal stresses in the sand on the non-expansion actuators and head expansion actuator during gait @5, respectively. μ signifies the coefficient of friction between DASR and sand, μ=tanφ, and φ stands for the friction angle between silica gel and sand.

The input air pressure p_{target} of the two expansion actuators is the same in this study, that is, $p_{target,\,2}$=$p_{target,\,5}$=p_{target}. Under a certain input air pressure p_{target}, $L_{d2,\,m}$=mL_{d2} and $D_{p,\,m}$=nD_p. On this basis, the following equations can be obtained by substituting Eqs. (8)–(11) into Eqs. (6) and (7):

$$\frac{\pi D_p^2}{4}\cdot q_{top,\,2}+\mu\pi D_p(L_{d1}+L_{d2})\cdot p_{2h}+G=\mu\pi nD_pL_{d3}\cdot p_{target}+\frac{\pi B^2D_p^2}{4}\cdot q_{t,\,2}, \tag{12}$$

$$\frac{\pi D_{p,\,m}^2}{4}\cdot q_{top,\,5}+\mu\pi D_p(L_{d3}+mL_{d2})\cdot p_{5t}+G=\mu\pi nD_pL_{d1}\cdot p_{target}+\frac{\pi D_p^2}{4}\cdot q_{t,\,5}. \tag{13}$$

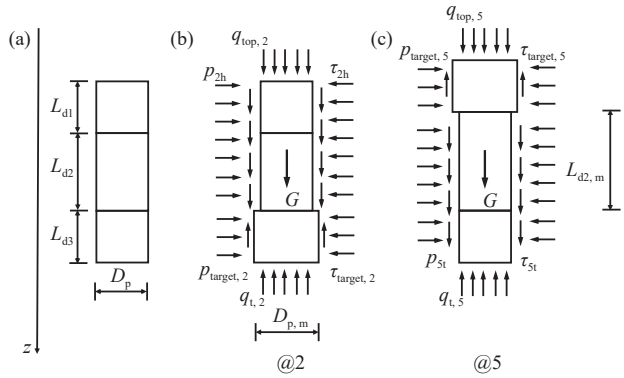

Fig.10 Schematic of the method for determining the geometric parameters of DASR. (a), initial condition; (b), mechanical balance at gait @2; (c), mechanical balance at gait @5.

The following equation exists between the total length (L_t) of DASR and L_{d1}, L_{d2}, and L_{d3}:

$$L_{d1} + L_{d2} + L_{d3} = L_t. \tag{14}$$

Similar to CPT and DMT, $q_{top,\,2}$, $q_{t,\,2}$, $q_{top,\,5}$, $q_{t,\,5}$, p_{target}, m, and n can be determined using cavity expansion methods (Yu, 2006, 2000; Huang and Tao, 2020; Martinez *et al.*, 2020). Then, we set L_t and D_p, and L_{d1}, L_{d2}, and L_{d3} can be determined according to Eqs. (12)–(14).

3.1.3 Assemblage

The fiber of the expansion actuator was at an angle of ±20° relative to the axial lines, and the fiber of the extension actuator was at an angle of ±85° (Fig.11a). In this study, L_t, D_p, and t of DASR were 220, 24, and 4.5 mm, respectively. n was regarded as 130% based on typical results from field tests (Yu *et al.*, 1996; Hoopes and Hughes, 2014). According to Eqs. (12)–(14), Section 2.5, and cavity expansion methods, the target pressure p_t (p_{target}) was considered to be 130 kPa. L_{d1}, L_{d2}, and L_{d3} were determined as 72, 60, and 72 mm, respectively. For DASR, the length-to-diameter ratios $\lambda_{da,\,exp}$ and $\lambda_{da,\,ext}$ of the expansion actuator and the extension actuator were 3.0 and 2.5, respectively. The D_p, L_t, and t of ECSR were 24, 220, and 4.5 mm, respectively. For ECSR, the length-to-diameter ratio $\lambda_{ec,\,ext}$ of the extension actuator was 8.8 (Fig.11b). The thicknesses of the inner and outer silicone layers of DASR and ECSR were 2.5 and 2.0 mm, respectively. The ends of the two soft robots were sealed with Dragon Skin 30, and air tubes were inserted at one of the ends. The three actuators of DASR were connected to each other with Dragon Skin 30, and holes were pre-drilled in the connecting silicone for the installation of air tubes. Silicone adhesive was utilized to seal the joints between the actuators and the air tubes. For the two soft robots, each actuator was pressurized using an individual air tube.

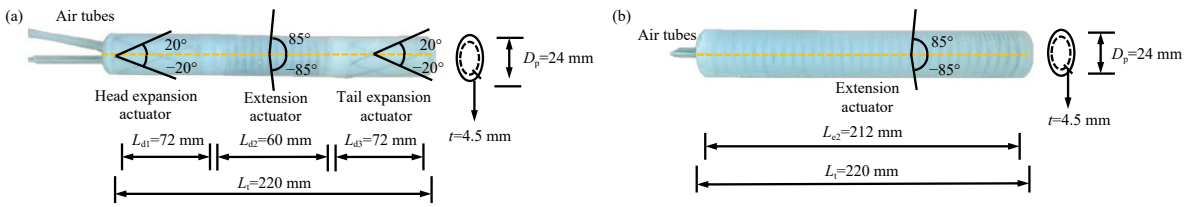

Fig.11 Assemblage of the two soft robots: dual-anchor soft robot (a) and extension-contraction soft robot (b).

3.2 Setup of the Burrowing-Out Test

As shown in Fig.12, the burrowing-out tests were conducted in a stainless-steel cylindrical barrel with an inner diameter (D_{chi}) of 390 mm, an outer diameter (D_{cho}) of 400 mm, and a height (H_{ch}) of 600 mm. A 50 cm scale was stuck to the inside of the barrel to measure the height of the sand sample. The stainless-steel barrel was placed on the vibrating table to vibrate sand to the target relative densities. A steel tube was adopted as the air input tube for the extension actuator, and a 40 cm measuring scale was attached to the steel tube to determine the depth of the soft robot during the tests. A steel frame was fixed around the barrel and vibrating table to ease observation and data recording. The burrowing-out process of the soft robots was captured using a camera with a rate of 24 frames per second.

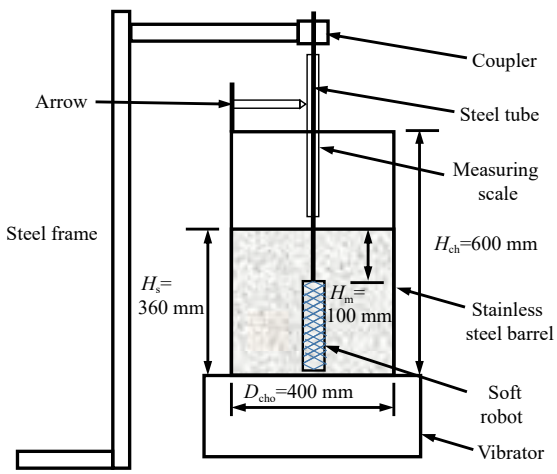

Fig.12 Setup of the burrowing-out tests.

The sand applied for the burrowing-out tests was the same as that described in Section 2.5, and the target relative densities (D_r) for the tests were 30%, 50%, and 80% (Table 1). The soft robot was buried in the sand for testing. The height (H_s) of the sand was 360 mm, and the distance from the top of the robot to the sand surface was 100 mm (Fig.12).

3.3 Typical Burrowing-Out Performance

The locomotion curve and the air pressure-time curve

Table 1 Testing cases for the burrowing-out test of the dual-anchor and extension-contraction soft robots

Case	Burrowing-out strategies	Relative density of sand (D_r)
A-D1	Dual-anchor	30%
A-D2	Dual-anchor	50%
A-D3	Dual-anchor	80%
E-D1	Extension-contraction	30%
E-D2	Extension-contraction	50%
E-D3	Extension-contraction	80%

of A-D2 and E-D2 for the first cycle are illustrated in Fig.13. Similar to our previous research (He *et al.*, 2024), the time each robot reached the final displacement in every gait lags behind p_t due to the hysteresis characteristics of the pneumatic actuators. The advancements of DASR and ECSR occurred when the extension actuators were activated. When the tail and head expansion actuators of DASR were activated, the locomotion curve exhibited two distinct horizontal lines, which indicate the anchoring phases. When the extension actuators were inactivated, both robots presented a certain downward segment in their displacement curves. This phase was called slip. For DASR, greater slip occurred when the head expansion actuator was inactivated. The reason could be that, when the extension actuator inactivated and contracted, the center of the extension actuator produced drag forces on both ends. The head expansion actuator could not generate sufficient anchoring force to balance the downward force and the slip. After the deflation of the head expansion actuator, the robot might also slip slightly under the action of gravity. The difference between advancement and slip was called a stride in a cycle.

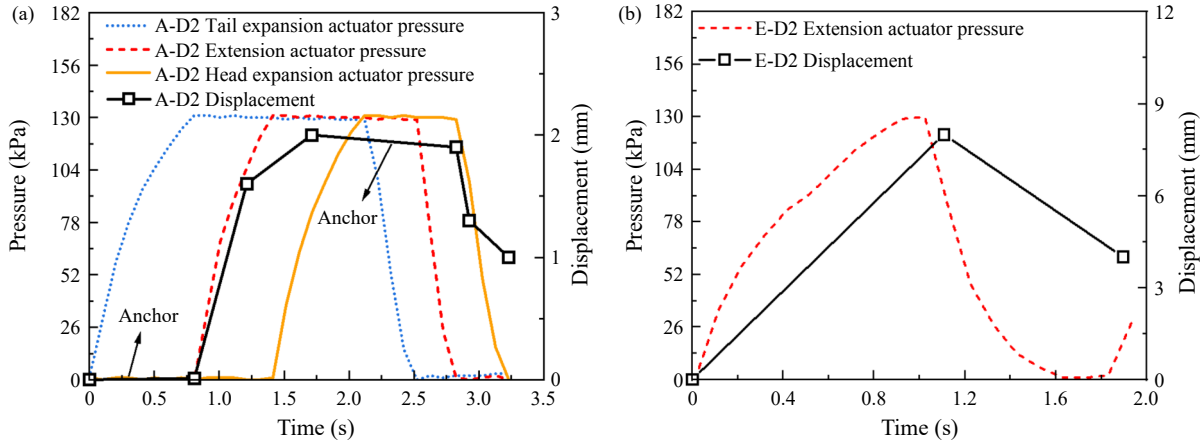

Fig.13 Air pressure and displacement of the first cycle. (a), dual-anchor soft robot; (b), extension-contraction soft robot.

Figure 14 shows the burrowing-out curves of A-D2 and E-D2. Figure 14a is an updated version of our previous research (He *et al.*, 2024). The curves existed in two stages: a gentle increase in the initial start-up stage and a rapid rise in the later fast motion stage. This phenomenon is similar to that observed by Tao *et al.* (2019, 2020) at T1D3 (D_r=69%). During the initial start-up stage, the cyclic motion of the robots sheared the sand, which led to a decline in the stress of sand. In this stage, the sand provided greater constraints on the robots, which resulted in a slower burrowing-out rate. When the sand stress weakened to a critical value, the burrowing-out curve started to enter the rapid improvement stage. Considering the anchoring of the DASR, the shear force acting on the sand in each cycle was greater than that for ECSR, which led to a smoother transition between the two stages of the motion curve. However, the rapid rise stage of ECSR appeared suddenly at a certain cycle. The stress on the

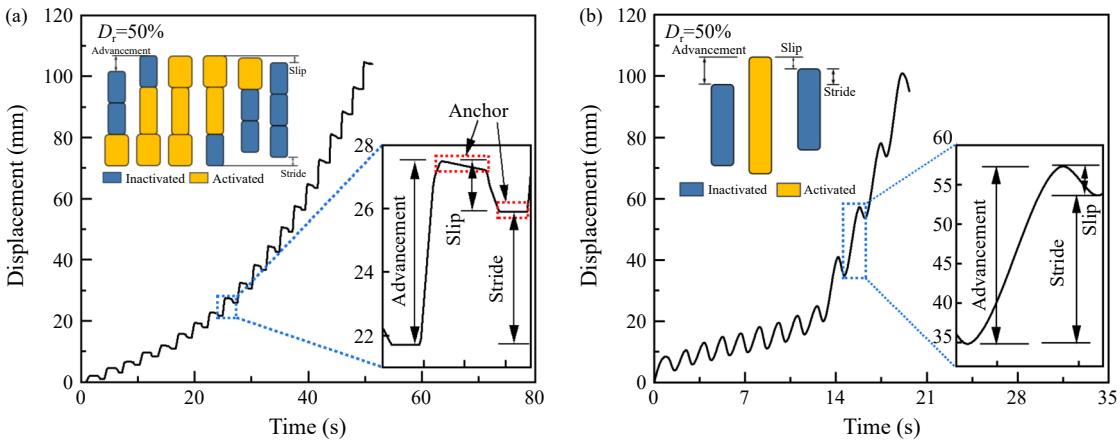

Fig.14 Typical burrowing-out curve of the soft robots. (a), dual-anchor soft robot (update from He *et al.* (2024)); (b) extension-contraction soft robot.

DASR weakened as the robot position rose. Thus, the anchorage generated by the expansion actuators gradually diminished.

This research used pressure control, whereas Tao *et al.* (2020) utilized time control. Consequently, the curve shapes of advancement, stride, and slip for ECSR were different from those observed by Tao *et al.* (2020). When D_r=50%, ECSR had a greater advancement than DASR within all cycles (Fig.15a). During the initial start-up stage, the advancement of ECSR in each cycle was maintained at approximately 8 mm. By the 10th cycle, a sudden increase was observed in the advancement of ECSR. The advancement of DASR rose linearly during all cycles. In all cycles, the slips of DASR were small (<2 mm), while the slips of ECSR were greater than 3.5 mm (Fig.15c). In the initial six cycles, the pressure acting on ECSR had not yet been decreased by shearing to a level that would allow for movement. This phenomenon resulted in a gradual rise in slip and a decline in stride (Fig.15b). When the strength of the sand weakened to the critical value, the slip began to gradually decrease, and the stride started to increase. Once a portion of ECSR penetrated beyond the sand, the anchorage performance reduced, which led to a further decline in slip. However, the local measurement of shear stress around the robots is difficult to determine. The abovementioned analysis may require additional validation in future works. Corresponding to the advancement, the stride of ECSR suddenly rose in the 10th cycle. Meanwhile, the stride of DASR improved linearly with the number of cycles. Evidently, DASR produces smaller slips in each cycle than ECSR, which demonstrates a more stable performance of burrowing-out.

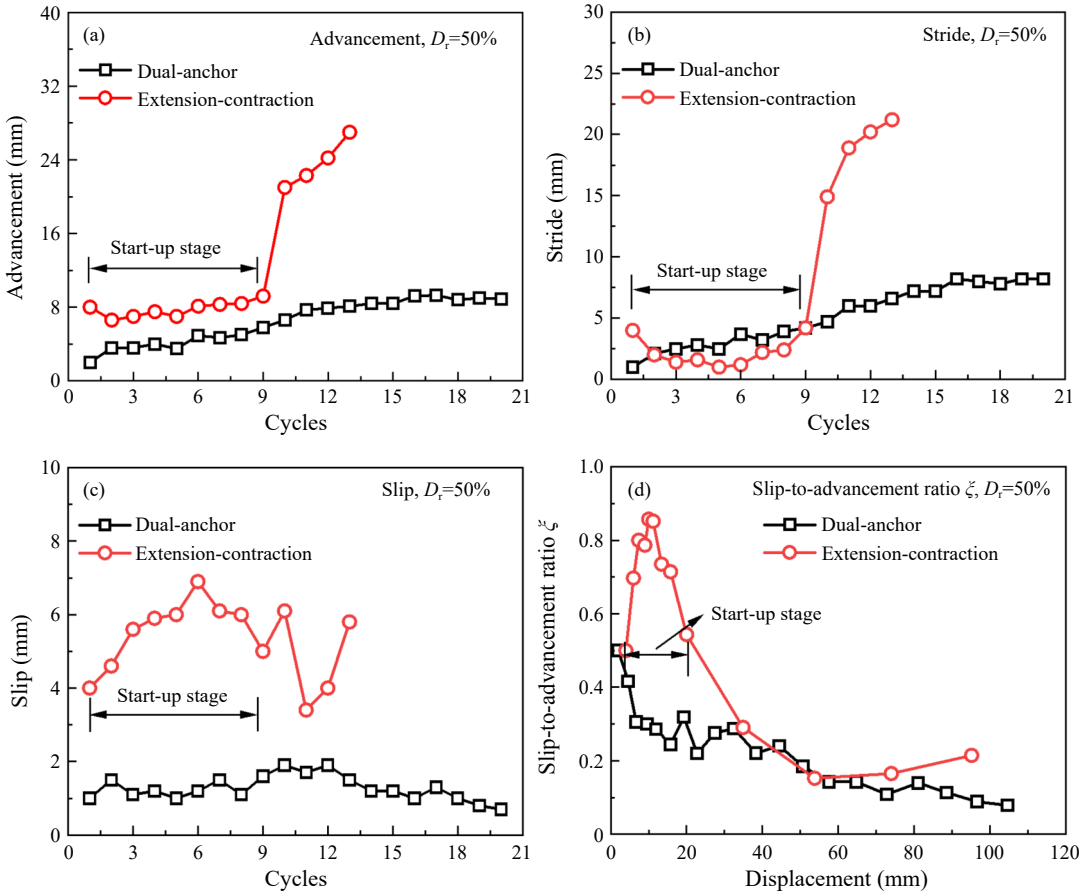

Fig.15 Burrowing-out performance of the soft robots at D_r=50%. (a), advancement; (b), stride; (c), slip; (d), slip-to-advancement ratio ξ.

To compare the resistance to slipping of DASR and ECSR during the penetration process, we defined the slip ratio ξ (<1) as follows:

$$\xi = \frac{d_{\text{slip}}}{d_{\text{advancement}}}, \tag{15}$$

where d_{slip} and $d_{\text{advancement}}$ denote the slip and advancement in each cycle, respectively.

Fig.15d presents the ξ values of DASR and ECSR during the burrowing-out process. Given that the total number of cycles for DASR and ECSR differed, the horizontal coordinates utilized the displacement values instead of the number of cycles to compare ξ between DASR and ECSR at the same position. During the entire burrowing-out process, the ξ of DASR gradually decreased and was lower than that of ECSR. Corresponding to the trend in slip changes, the ξ of ECSR initially rose, then declined, and ultimately presented a slight upward trend. This variation in ξ further indicates that ECSR nearly did not exhibit any movement during the initial stage. DASR demonstrated a stronger capability to resist slipping than ECSR.

3.4 Burrowing-Out Performance in Different Sand Densities

Figure 16 presents the cycle numbers and burrowing-out curve of DASR and ECSR for burrowing-out in different sand relative densities D_r. As D_r rose, the strength and stiffness of the sand improved, which required the robots to undergo more cycles to complete the initial start-up stage. The robots spent more time in the initial start-up stage with rising relative densities. ECSR utilized fewer cycles to complete the burrowing-out process in low and medium relative density sands (D_r=30% and D_r=50%). However, ECSR needed more time to complete the initial start-up stage in sand of high relative density (D_r=80%) (Fig.16b). Thus, it required a higher number of cycles to burrow out (Fig.16a). ECSR started faster than DASR at D_r=30% and D_r=50%. However, ECSR took twice as much time as DASR to complete the initial start-up stage at D_r=80%, and the number of cycles applied for burrowing out was four times that of the DASR. The burrowing-out performance of ECSR sharply declined with an increment in relative density.

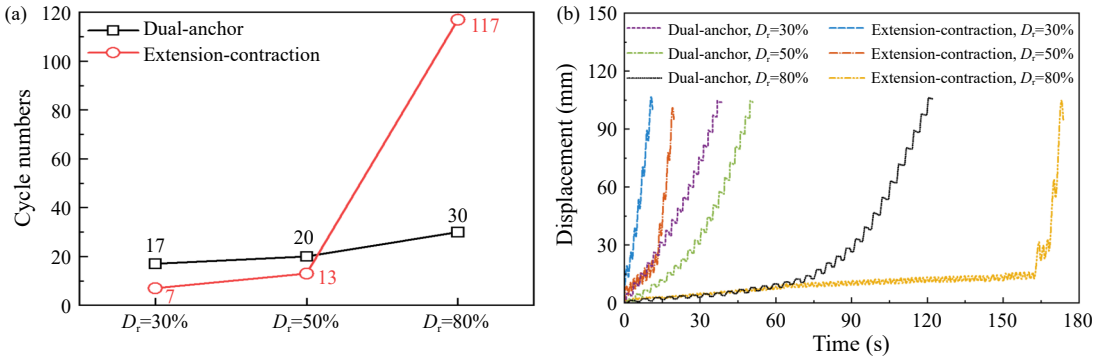

Fig.16 Burrowing-out performance of the soft robots at different sand densities. (a), cycle numbers for burrowing-out; (b), burrowing-out curve.

Figure 17 displays the burrowing-out performance of DASR and ECSR at different sand relative densities. The advancement of DASR was lower than that of ECSR in sands with medium and low relative densities. However,

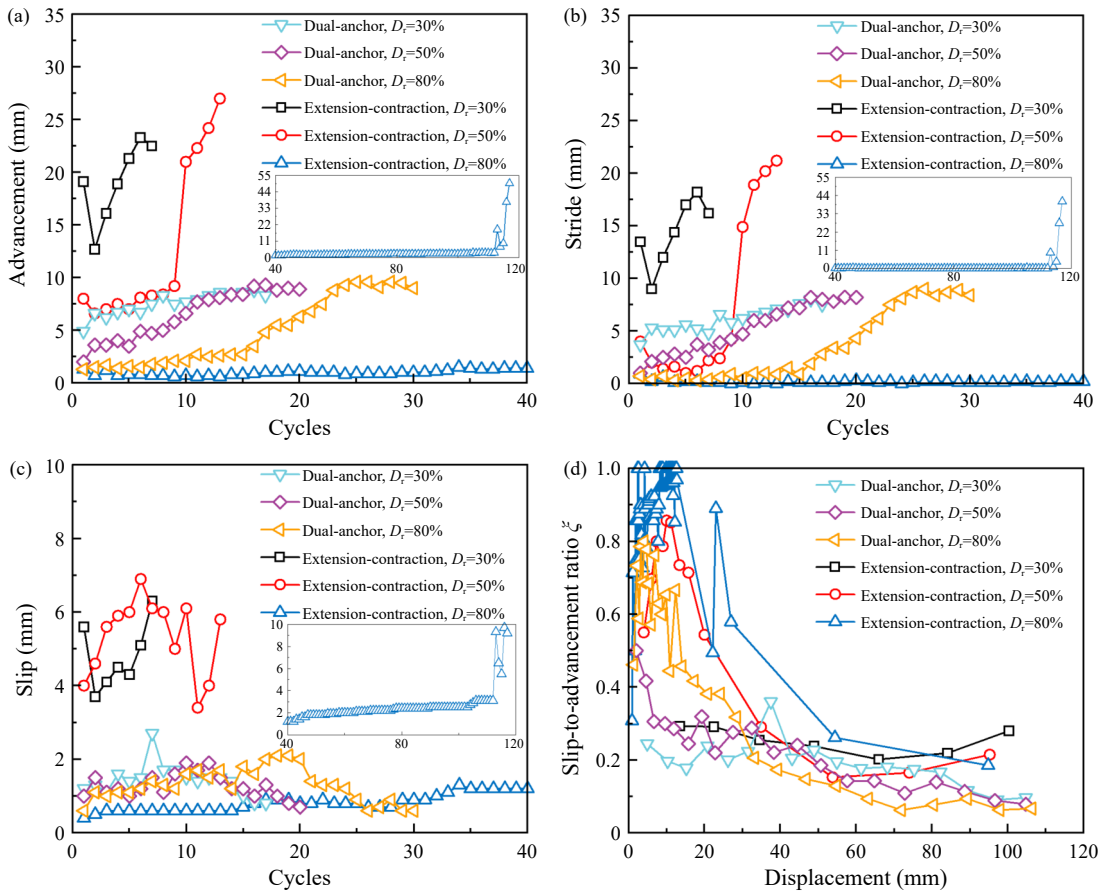

Fig.17 Burrowing-out performance of the soft robots at different sand densities. (a), advancement; (b), stride; (c), slip; (d), slip-to-advancement ratio ξ.

in sand with high relative density, the advancement of DASR significantly improved, which demonstrates its effective burrowing-out performance in dense sand (Fig.17a). The advancement of ECSR declined with rising relative density. At D_r=80%, nearly no advancement occurred until a sudden increment in the final few cycles. The strides of DASR and ECSR were similar to their advancement. The stride of ECSR initially remained within a lower range before a sudden rise. The stride of DASR steadily rose in sands with medium and low relative densities. In dense sand, it started at lower values and began to gradually improve (Fig.17b). The slip of DASR remained roughly consistent across different relative densities. DASR experienced less slip than ECSR in sands with medium and low relative densities. In dense sand, ECSR needed more cycles to initiate movement. Within the number of cycles for DASR, ECSR nearly did not move, which resulted in less slip than that of DASR (Fig.17c). The ξ curve of the robots presented two distinct patterns: it gradually decreased with increasing cycles in low relative densities (pattern I) and initially rose before gradually declining in high relative densities (pattern II). A critical relative density ($D_{r,\,lim}$) existed, at which the ξ curve of the robot switched between the two patterns. $D_{r,\,lim}$ for DASR was higher than that for ECSR. DASR demonstrated superior burrowing-out performance in dense sand, with higher advancement and stride, as well as lower slip and ξ. This performance can be attributed to the dual-anchor strategy, which actively generates stable anchoring points at both ends. Ultimately, the impact of external disturbances is reduced, and slip is minimized. As shown in Fig.15d, the slip-to-advancement ratio of DASR is consistently lower than that of ECSR, which confirms its stability advantage. By contrast, ECSR relies on the natural vertical stress gradient for anchoring, which is less effective in dense sand and thus leads to increased slip and instability. Therefore, the dual-anchor strategy is more suitable for locomotion in dense sand, while the extension-contraction strategy might perform better in looser sand.

Figure 18 shows the maximum value ξ_{max}, minimum value ξ_{min}, and mean value ξ_{mea} of the robots at different sand relative densities. ξ_{max} and ξ_{mea} rose with relative density, while ξ_{min} changed slightly. ξ_{max}, ξ_{min}, and ξ_{mea} for DASR were lower than those for ECSR across all relative densities. In loose sand, the difference in ξ_{mea} between DASR and ECSR was small, but this difference became pronounced as relative density improved. This finding indicates that DASR has superior burrowing-out performance to ECSR in terms of advancement efficiency.

3.5 Burrowing-Out Efficiency in Different Sand Densities

Burrowing-out efficiency refers to the effectiveness of the robot in realizing upward motion relative to the energy input and cycle count. In this work, three primary indicators were utilized to analyze efficiency: burrowing-out average velocity $\bar{v}$, total energy consumption W_{total}, and COT. $\bar{v}$ could be calculated using the following equation:

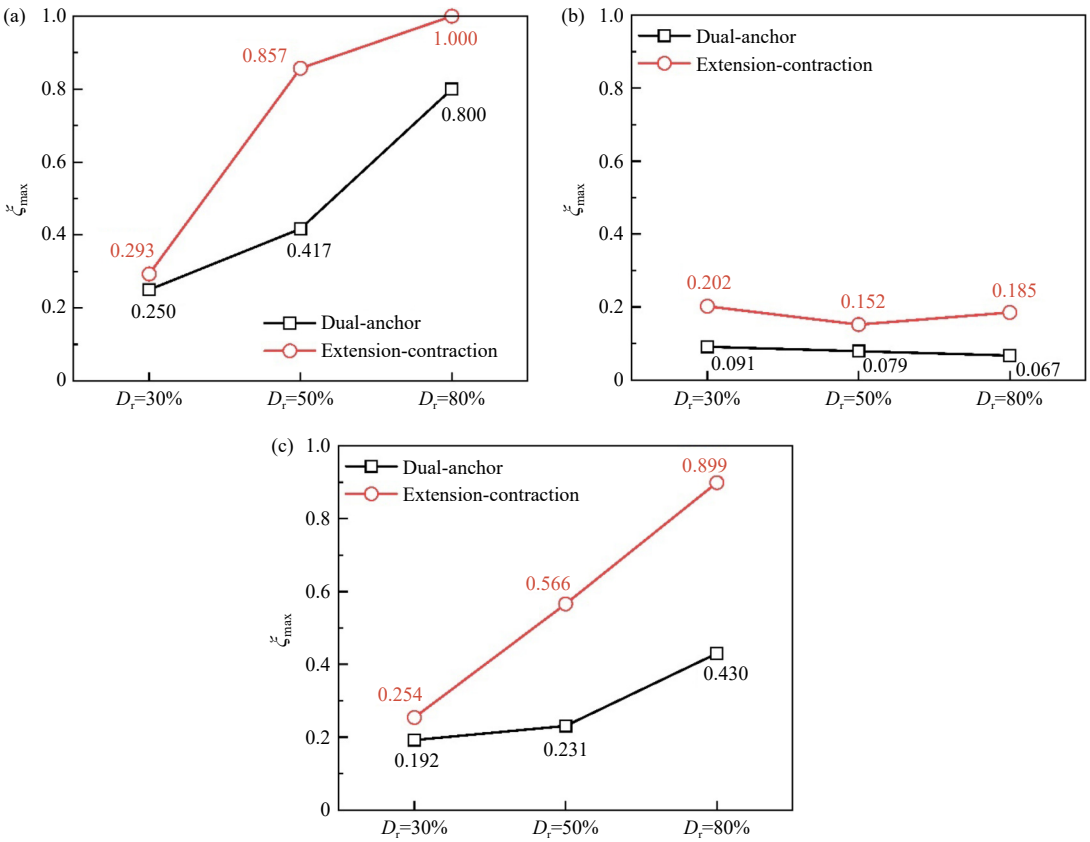

Fig.18 Slip-to-advancement ratio ξ at different relative densities. (a), maximum value ξ_{max}; (b), minimum value ξ_{min}; (c), mean value ξ_{mea}.

$$\bar{v} = \frac{s}{t_{\text{total}}}, \tag{16}$$

where $\bar{v}$ is the average velocity of the robots throughout the burrowing-out process (mm/s). s denotes the total displacement at the end of the final cycle of the burrowing-out process (mm). t_{total} signifies the total time to complete the burrowing-out process (excluding the intervals between inflation and deflation).

Considering the negligible power consumption of the Arduino board compared with that of the air pump and solenoid valves, W_{total} was considered the sum of the energy consumed by the air pump and solenoid valves. A displacement modification factor (η) was introduced to reduce the impact of varying s across different cases for facilitating comparison. The calculation of W_{total} is as follows:

$$W_{\text{total}} = \eta W_{\text{m}}, \tag{17}$$

$$\eta = \frac{s_0}{s}, \tag{18}$$

$$W_{\text{m}} = W_{\text{pump}} + W_{\text{valve}}, \tag{19}$$

where W_{m} refers to unmodified total energy consumption (J). s_0 is target displacement for modification (s_0=100 mm). W_{pump} and W_{valve} denote the energy consumption of the air pump and solenoid valves during the burrowing-out process, respectively (J).

The air pump was operated only during inflation. The solenoid valves were in operation except when they were in a sealed state to hold air pressure. Calculations revealed that the energy consumption for inflating and deflating with three solenoid valves operating in predetermined cycles aligned with that of a single solenoid valve. The energy consumption of the air pump and solenoid valves was computed based on actual running time and power. The calculation method is as follows:

$$W_{\text{pump}} = P_{\text{pump}} t_{\text{inf}}, \tag{20}$$

$$W_{\text{valve}} = P_{\text{valve}} t_{\text{total}}, \tag{21}$$

$$t_{\text{total}} = t_{\text{inf}} + t_{\text{def}}, \tag{22}$$

where P_{pump} and P_{valve} denote the power of the air pump and solenoid valves during the burrowing-out process, respectively (W). t_{inf} and t_{def} signify the total inflation time and total deflation time during the burrowing-out process of the robots, respectively (s).

COT is widely utilized to assess the locomotion efficiency of robots or organisms, with lower values indicating higher efficiency. It could be calculated as follows:

$$COT = \frac{W_{\text{m}}}{m \cdot s}, \tag{23}$$

where m represents the mass of the robots. The m values of DASR and ECSR were 101.76 and 89.13 g, respectively.

The $\bar{v}$ of the robots decreased linearly with the increase in D_{r}, with ECSR having a steeper slope than DASR. The $\bar{v}$ of ECSR was higher than that of DASR in sands with medium and low relative densities (D_{r}=30% and D_{r}=50%), but it was lower in sand with high relative density (D_{r}=80%) (Fig.19a). This result indicates that dual-anchor enhances the burrowing-out performance of the robots in sand with high relative density. The robots needed more cycles to burrow out in sand with high relative density, which led to an increase in W_{total} as the relative density rose. In addition, ECSR required more cycles to complete the initial start-up stage in sand with high relative density, with its energy consumption being twice that of DASR (Fig.19b). This efficiency difference is primarily attributed to the anchoring strategy rather than the geometry of the actuator. The dual-anchor mechanism of DASR ensures that each expansion cycle results in effective advancement with minimal slip, which reduces unnecessary energy losses. By contrast, reliance on ECSR on the natural vertical stress gradient leads to inefficient anchoring, which results in excessive energy expenditure on slip correction. As presented in Fig.19c, DASR has significantly lower *COT* in sand with high relative density, which demonstrates its superior efficiency. Therefore, the anchoring strategy plays a crucial role in determining the energy performance of the robot.

3.6 Burrowing-Out Mechanism of Dual-Anchor and Extension-Contraction

The locomotion of finite-length objects initiates from the asymmetry in boundary conditions, and the degree of asymmetry reflects the effectiveness of burrowing-out. In subterranean locomotion, a robot must form temporary anchor points to counteract the resistance of granular media and advance efficiently. The deliberate amplification of this asymmetry is an important concept in the design of subterranean robots, which affects their capability to decrease slip and enhance energy efficiency.

The natural vertical stress gradient in the soil causes ECSR to passively form anchor points during inflation and deflation (Fig.20b). As the actuator expands, it presses against the surrounding sand, which generates frictional resistance that serves as a temporary anchor. However, this mechanism relies on local stress distributions, which makes the anchoring effect inherently unstable. This influence is specifically obvious in sand with high relative density, where stress gradients are less pronounced. As a result, ECSR presents a greater slip and needs more cycles to complete the burrowing process. The variation in slip observed in Figs.15(c)–(d) and Figs.17(c)–(d) suggests that the anchoring force of ECSR fluctuates, which leads to inconsistent advancement per cycle. In loose sand, the stress gradients are higher, which results in more effective passive anchoring, as reflected in higher advancement per cycle in Fig.17a. However, in dense sand, the weaker anchoring force leads to excessive slip, which requires more energy to correct unintended downward displacement (Fig.19b).

By contrast, DASR actively generates anchoring points at both ends through its expansion actuators while reinforcing asymmetry in boundary conditions (Fig.20a).

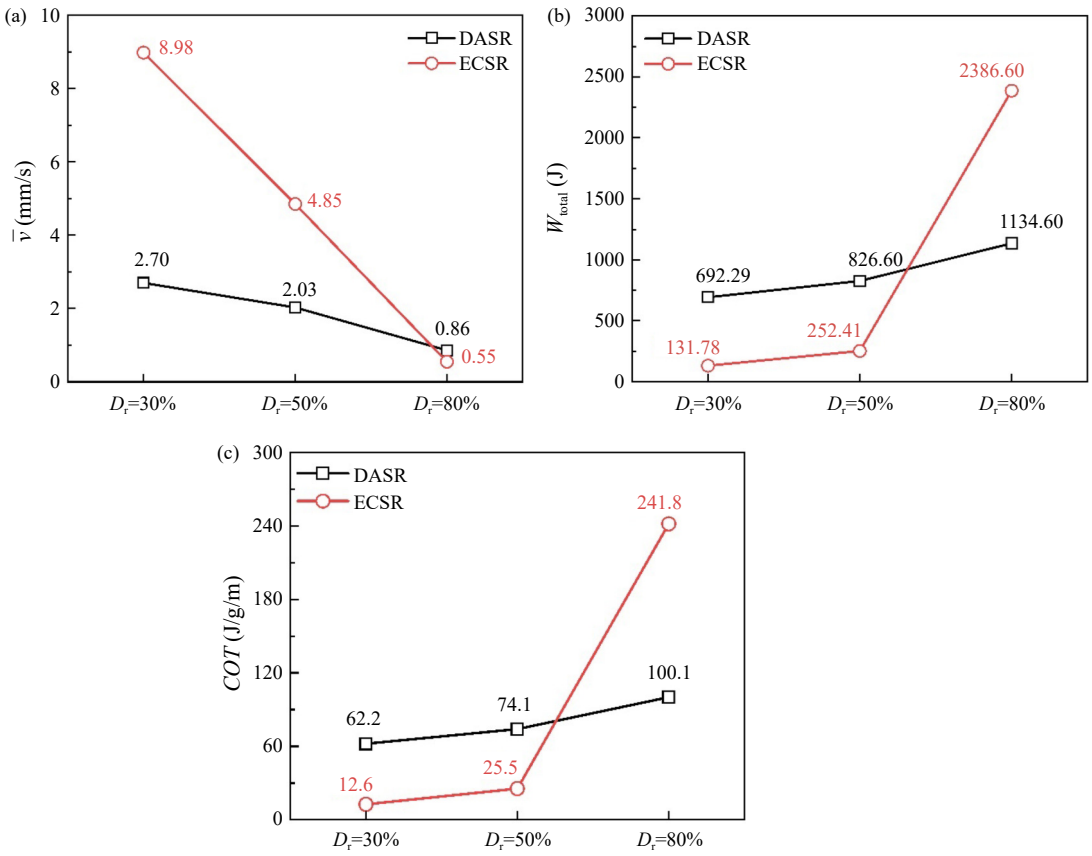

Fig.19 Average speed $\bar{v}$, total energy consumption W_{total}, and cost of transportation (*COT*) of the soft robots. (a), $\bar{v}$; (b), W_{total}; (c), *COT*.

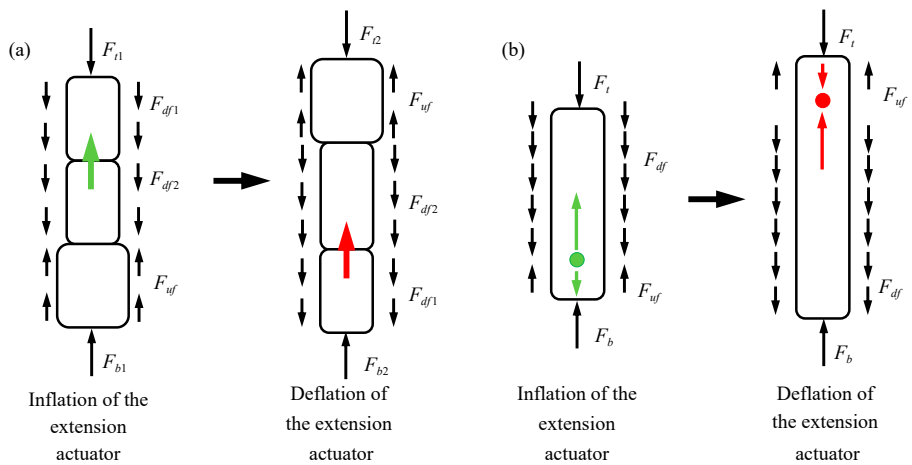

Fig.20 Burrowing-out mechanism of dual-anchor and extension-contraction. (a), dual-anchor; (b), extension-contraction.

When the tail expansion actuator inflates, it exerts a radial force against the surrounding sand, which significantly increases frictional resistance and forms a stable anchor. Then, the extension actuator extends to push the robot upward. Upon reaching the target position, the head expansion actuator inflates, which secures the new position, while the tail anchor releases to allow repositioning. This alternating anchoring strategy guarantees that each movement cycle produces consistent and controlled advancement, which reduces slip and improves energy efficiency. The effectiveness of this anchoring mechanism is validated by the lower slip-to-advancement ratio (ξ) in Figs.15d and 17d and the reduced *COT* in Fig.19c.

The stability of the anchoring mechanism directly affects energy consumption. Given that ECSR encounters more slip, a larger portion of energy is wasted on correcting downward displacement. This wastage leads to higher *COT* values. Conversely, DASR maintains a more stable anchoring mechanism, which allows each expansion cycle to contribute effectively to upward motion. Ultimately, unnecessary energy loss is decreased. Consequently, DASR demonstrates superior energy efficiency in sand with high relative density (Fig.19c).

The comparison between DASR and ECSR empha-

sizes the importance of active anchoring strategies for enhancing burrowing efficiency in granular media. While passive anchoring (ECSR) is effective in loose sand, it becomes unstable in dense environments. On the contrary, active anchoring (DASR) guarantees stable progression, which reduces slip and enhances energy efficiency. Thus, it is a more robust solution for subterranean locomotion in varying soil conditions.

4 Conclusions

In this study, a DASR and an ECSR inspired by the locomotion of razor clams were constructed and comparatively investigated. The following conclusions can be made:

1) The effects of input air pressure (p) and length-to-diameter ratios (λ) on the deformation of the fiber-reinforced pneumatic extension and expansion actuators were calibrated. The results revealed that an increase in p led to a rise in deformation of the actuators, and the optimal air pressure was 130 kPa. λ had little effect on the extension for the extension actuators. The expansion of the expansion actuators rose when $\lambda<2.5$ and tended to be the same for the range $\lambda\geq2.5$.

2) The effect of p and λ on the penetration force (F_p) for the extension actuators and on the anchorage force (F_a) for the expansion actuators were examined in different sand relative densities (D_r=30%, 50%, 80%). F_p increased with the rise in λ. F_a rose slightly in sand of different relative densities, with the exception that F_a improved abruptly as λ enhanced from 2.5 to 3.0 in dense sand.

3) We proposed an analytic method based on the force equilibrium to ascertain and adjust the geometric parameters of DASR. Based on this method, the lengths of the head, middle, and tail segments for the DASR were 75, 60, and 75 mm, respectively. The diameter of the segments was 24 mm.

4) Burrowing-out tests in dry sand with different relative densities were conducted. The results demonstrated that DASR had high efficiency in dense sand in terms of lower time of burrowing-out, slip-to-advancement ratio, and *COT*. In dense sand, DASR could effectively raise anchoring force to reduce slips and present higher burrowing-out efficiency. ECSR might perform better in looser sand in terms of higher burrowing-out average velocity, higher advancement in each cycle, and lower energy consumption. However, it had much larger slips than DASR. DASR could realize steady advancement and net displacement in each cycle and effectively decrease slips.

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Author Contributions

Xin Huang: writing–original draft, formal analysis, data curation. Jia He: writing–review and editing, methodology, conceptualization. Hao Wang: writing–review and editing, experiment. Fengyuan Yan: writing–review and editing. Hailei Kou: writing–review and editing.

Data Availability

All data generated and analyzed during this study are included in this published article and its additional files.

Declarations

Ethics Approval and Consent to Participate

This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for Publication

Informed consent for publication was obtained from all participants.

Conflict of Interests

The authors declare that they have no conflict of interests.

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